

Matias Freire

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EDUCATION

Stevens Institute Of Technology

Expected May 2028

B.S. in Computer Science

Hoboken, NJ

Awards: STEP Scholar: Selected from a pool of 200+ students based on outstanding leadership and commitment to the program.

Coursework: Algorithms and Data Structures, Computer Architecture, Systems Programming, Software Development Processes

Certifications: Bloomberg Market Concepts, Hugging Face AI Agents Fundamentals

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript

Web Tools: TypeScript, React, Tailwind, Next.js

Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks, Linux

Libraries: Dash, Pandas, NumPy, PyTorch, Scikit-learn, Streamlit

EXPERIENCE

Incoming Summer Software Engineering Intern

May 2026 – Aug 2026

Wabtec Corporation

Erie, PA

- Will work across embedded systems and modern software engineering, including C/C++ on Linux with Docker, as well as React, PyTorch, and AI development tools.

Deep Learning Research Assistant

Aug 2025 – Mar 2026

Advanced Quantum Materials Lab

Hoboken, NJ

- Developed a Dash-Plotly platform integrating literature-driven preprocessing, synthetic data, and open-source pipelines for interactive MoS defect visualization and DL training.
- Studied preprocessing methods, synthetic data generation, and deep learning models (CNNs like MobileNet), Extraordinary Magnetoresistance (EMR) devices and learned how to replicate it's magnoreistence results on COMSOL.

ML Ignite Fellow

Sept 2025 – Jan 2026

AI4ALL

Remote

- Completed a 10+ week machine learning program applying Python tools (Pandas, NumPy, Scikit-learn) to analyze datasets and build predictive models across multiple hands-on labs.
- Collaborated with a Capital One mentor to develop a final project, translating ML/DL theory into a working solution while receiving industry-oriented feedback on model selection and career direction.

PROJECTS

MoS2 Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly, CSS*

[Github-Repo](#)

- Designed and built from scratch an end-to-end Dash-Plotly research platform for deep learning-based MoS defect analysis
- Automated synthetic image generation with configurable defects and realistic noise injection, integrating a ResUNet model for vacancy and polymorph classification

SHPE and Colorstack Stevens Chapter Website | *Next.js, React, Tailwind, Typescript*

[Github-Repo](#)

- Built from scratch a modular, responsive Next.js and React website, serving 60+ and 150+ members, with the goal of entry point to attract new sponsors and inform new members
- Established a reusable UI system through rapid HTML/CSS prototyping and full implementation in TypeScript and Tailwind CSS

Interactive-Neural-Network-Digit-Classifer | *Python, Pytorch, Pygame, Streamlit*

[Github-Repo](#)

- Developed a Streamlit application for real-time handwritten digit recognition using a neural network achieving 96% test accuracy, Integrated a Pygame-based drawing canvas enabling users to input custom digits for live model inference

LEADERSHIP

Chief Web Officer | *SHPE Stevens Chapter*

March 2026 – Present

- Leading development of the official chapter website in Next.js, building the platform from scratch to support 200+ registered members and present the organization to professional sponsors
- Planning and organizing technology-focused events and workshops to expand technical engagement within the chapter

Vice-President - Previous Head of External Relations | *Colorstack Stevens Chapter*

April 2025 – Present

- Led growth and technical engagement initiatives, increasing active membership by 50% and online engagement by 25% through social media strategy and 14+ events per semester
- Organized peer-led technical sessions, guiding members through 32+ data structures problems

STEP Tutor-Counselor/Resident Assistant (TCRA) | *STEP Bridge Summer Program*

Jun 2025 – Aug 2025

- Led Calculus I/II and Physics recitations for 52+ incoming freshmen, strengthening STEM readiness through structured reviews and weekly after-hours tutoring for 25+ students
- Served as a Resident Assistant, personally mentoring and tutoring 52+ residents while enforcing residential policies and supporting academic success, wellness, and community standards